Occlusion-Aware Road Extraction Network for High-Resolution Remote Sensing Imagery

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***Abstract*— Road occlusion seriously affects the connectivity of extracted roads, and has a negative effect on practical applications. The dense road occlusion problem is caused by high-rise buildings and street trees, and is a more serious and unique problem than simple occlusion caused by low buildings and scattered trees. The existing methods mainly solve the road occlusion problem by enhancing the encoder ability to capture the long connectivity feature of roads. Unfortunately, the existing methods can only solve small and sparse road occlusion situations, and they cannot deal with the dense road occlusions caused by dense high-rise buildings or trees. In this article, to solve the dense road occlusion problem, the occlusion-aware road extraction network, namely OARENet, is proposed for road extraction from high-resolution remote sensing imagery. In OARENet, an occlusion-aware decoder (OADecoder) is designed by explicit modeling the texture feature for road regions with dense occlusions. The OADecoder is made up of a regular occlusion-aware (ROA) module and a stochastic occlusion-aware (SOA) module. The ROA module is implemented by adopting different dilation rates to fit the texture feature in the semantic feature maps. The SOA module is proposed by designing stochastic convolutions to adaptively fit the spatial details of road regions with dense occlusions. In order to evaluate the dense occlusion problem, a dense occlusion road dataset (JHWV) was built and annotated. The experimental results obtained on the DeepGlobe dataset, the newly built JHWV dataset, and large-scale urban images demonstrate the superiority of OARENet, especially when faced with a dense road occlusion situation. Code has been made available at: https://github.com/WanderRainy/OARENet.**

***Index Terms*— Dense road occlusion, occlusion-aware, remote sensing images, road extraction, semantic segmentation.**

I. Introduction

AUTOMATICALLY extracting roads from high-resolution remote sensing is an important task for many applications, such as urban planning [1], [2], updating geographic information systems (GIS) [3], [4], automatic driving [5], [6], and other application fields [7], [8]. The road data acquisition method of field measurement cannot easily meet the needs of large-scale and real-time road data in all walks of life, which is exactly what high-resolution remote sensing imagery offers [9], [10]. Therefore, road extraction from high-resolution remote sensing imagery has drawn widespread attention [11], [12], [13].

Compared to the traditional algorithms that focus on hand-crafted features to extract roads [14], [15], [16], deep neural network models [17], [18], [19] have achieved a satisfactory accuracy in the automatic extraction of roads from remote sensing imagery [20], [21], especially with the open road dataset [7], [22], [23]. Road occlusion by buildings and trees is the most serious problem for road extraction, which causes the extracted roads to be fragmented. The existing methods mainly capture the long-distance connectivity feature of roads to solve the road occlusion problem, which can achieve a satisfactory performance with small and sparse occlusions [24], [25]. However, when high-resolution remote sensing imagery contains dense high-rise buildings or trees, the prior approaches do not handle these dense occlusions well.

In practical applications in Chinese cities, dense occlusions frequently appear in high-resolution remote sensing imagery and can cover most of the road surface. Dense road occlusion situations are shown in Fig. 1, where continuous high-rise buildings and trees surround the roads, which causes many areas of the road surface to be obscured by the buildings and trees in the high-resolution remote sensing imagery. Dense road occlusions form many holes in a strip of road from the perspective of geometry, which results in the road shape not being a continuous strip, but instead many fragments. The existing road extraction methods are based on a fundamental hypothesis about road features from high-resolution remote sensing imagery, which is that roads in remote sensing imagery are strip-shaped objects whose intensity and width do not change dramatically [26]. However, this hypothesis is not suitable for roads with dense occlusions, the surface of which has complex texture changes in remote sensing imagery. Road regions with dense occlusions are composed of occlusion objects with complex texture and road surfaces with a smooth texture. Therefore, compared to a simple road situation, road regions with dense occlusions have a unique texture feature, which is that the local smooth texture is distributed discretely on the strip. The reason for the universal models of linear targets performing poorly in road regions with dense occlusions is the unique texture feature, which differs from that of a simple road situation. As shown in Fig. 1, when the existing methods are applied to large-scale remote sensing satellite images of cities with dense high-rise buildings or trees, a mass

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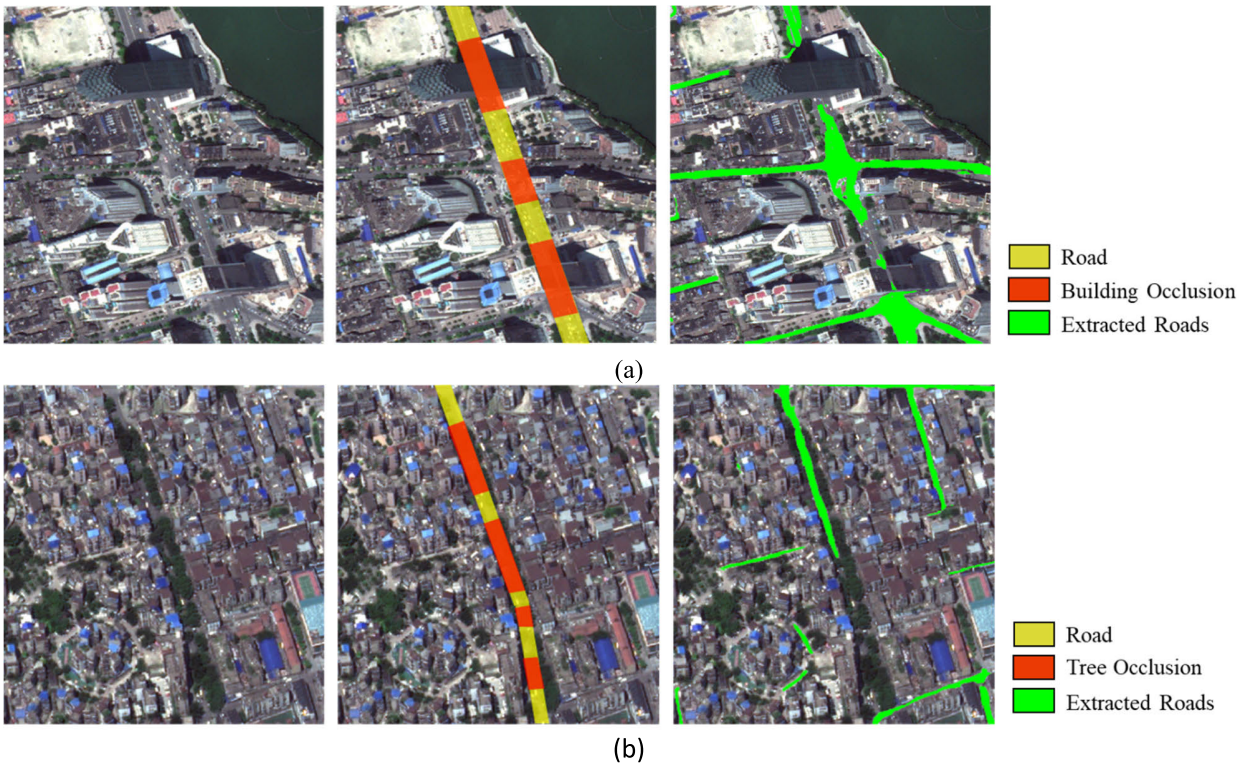


Fig. 1. Motivation for the proposed occlusion-aware decoder. Dense occlusions destroy the long connectivity feature of roads. The extracted roads of the state-of-the-art model [25] are fragmented. (a) Dense road occlusion by high-rise buildings. (b) Dense road occlusion by trees.

of road fragments can appear in the extracted results, which leads to difficulties in the application of high-resolution remote sensing imagery.

In this article, to address the above problems, the occlusion-aware road extraction network (OARENet) is proposed to solve the dense road occlusion problem for high-resolution remote sensing imagery. Due to dense road occlusions destroying the clear long connectivity feature of roads, it is difficult for the encoder to capture sufficient road features. Moreover, the decoder recovers detailed spatial features, and it is difficult for the normal convolution operation to avoid recovering occlusion information. However, roads with dense occlusions have a special shape feature, which is a continuous strip with holes, so strip convolutions with holes are more aligned with the shape feature of a road with occlusions. Explicit modeling based on strip convolutions [27] is more reliable for the unique texture feature of road regions with dense occlusions. Based on the above, the occlusion-aware decoder (OADecoder) is proposed.

The major contributions of this research are summarized as follows.

1) An OARENet is proposed to solve the dense road occlusion problem. Considering the unique texture of road regions with dense occlusions, the OADecoder is designed by explicit modeling the texture feature to enhance the awareness of road regions with dense occlusions, based on strip convolutions. The OADecoder consists of a regular occlusion-aware (ROA) module and a stochastic occlusion-aware (SOA) module, which are designed for the different hierarchical feature maps from high-resolution remote sensing imagery.
2) The occlusion-aware module is designed by explicit modeling the texture feature to increase the awareness of road regions with dense occlusions, instead of only roads. To avoid occlusion information being recovered in the upsampling, the occlusion-aware module is introduced in each decoder block after upsampling and skip connection. The ROA module adopts different dilation rates in the semantic feature maps, and uses multiscale channels to fuse the features of different scales. The SOA module is designed to fit the spatial details of road occlusions with any situation generated by random sequences.
3) A new dense occlusion road dataset (JHWV) was built and annotated to evaluate the dense road occlusion problem. Compared to the existing road datasets, the JHWV dataset includes road regions with the dense occlusion problem, to make up for the shortcomings of the current road datasets in practical applications. The proposed OARENet method was found to achieve better connectivity when faced with roads with dense occlusions, and can be applied with different encoders to obtain better road extraction results.

The rest of this article is structured as follows. In Section II, the related works on road occlusion are presented. In Section III, we introduce the details of the proposed OARENet method. In Section IV, the datasets and implementation details of the experiments are provided, along with the experimental results. Section V provides an analysis of the proposed OADecoder on high-resolution remote sensing imagery from cities with dense occlusions. Our conclusions are presented in Section VI.

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II. Related Work

Over the past few decades, in order to solve the occlusion problem, many approaches have been proposed from different viewpoints to refine the extracted road networks. These approaches can be roughly divided into three main categories: 1) methods based on postprocessing, 2) methods based on the direct generation of vector maps, and 3) methods based on optimization of the model structure [26], [28].

Introducing a postprocessing step based on the initial road segmentation map can directly improve the road extraction results, particularly when the road extraction model fails to predict road regions with occlusions. A fully connected conditional random field (CRF) model [29] was proposed to refine the road results extracted by the segmentation model [30]. An end-to-end road vectorization mapping network framework [31] was proposed to use nodes based on a road refinement module, to improve the road connectivity. The A* search algorithm was used to generate connection hypotheses to solve the missing connections in road networks [32]. These methods can refine road networks, but cannot solve complex road occlusion environments, and are limited by the ability of the road segmentation model.

The topological structure is an important feature of roads [33], and researchers have proposed many algorithms that can directly generate vector roads to obtain a road map with better connectivity. The RoadTracer method was proposed by [34] to automatically construct road network maps from aerial images, which uses an iterative search method to derive the road network. PolyMapper [35] is a sequentialization method based on Polygon-RNN [36], which reformulates a graph structure as closed polygons to fit the shape of roads. Tan et al. [37] proposed VecRoad, which is a point-based iterative graph exploration scheme that can achieve better alignment of roads. Although the road extraction results obtained by the above methods typically have good connectivity, the road surface generated by relying on the extracted road centerline can be very different from the actual road surface, so that the accuracy of the obtained road regions is not as good as the results obtained by directly using a segmentation method.

Recently, many researchers have paid attention to improving the model structure by optimizing the ability to capture long-distance features to extract roads. In the CVPR DeepGlobe 2018 Road Extraction Challenge [7], the D-LinkNet method [8] was proposed, which adopts dilated convolutions to enlarge the receptive field for road extraction based on LinkNet [38]. Atrous spatial pyramid pooling has also been proposed to improve the performance of road extraction by enhancing the context awareness [39], [40]. DBRANet [41] introduces an improved encoder–decoder method, which improves the fusion of feature maps of different scales. The GAMSNet method, which was proposed by [25], uses multiscale and global awareness, capturing the spatial context to improve the road recognition performance. CoANet [42] uses strip convolutions to capture long-range context information and explores the relationship between neighboring pixels for better connectivity. The road extraction results can be improved by the above methods, and some simple occlusion problems can be solved. However, dense occlusions caused by high-rise buildings or trees in cities destroy the long connectivity feature of roads, which causes the above methods to only extract fragmented road results.

III. Methodology

In this article, to solve the dense occlusion problem, OARENet is proposed for high-resolution remote sensing imagery, as shown in Fig. 2. In OARENet, the OADecoder is proposed to enhance the awareness of road occlusions, which consists of an ROA module and an SOA module. The ROA and SOA modules are designed to align with the texture feature of road regions with dense occlusions and are used in the different hierarchical feature maps.

A. Occlusion-Aware Road Extraction Network Structure

The model structure of OARENet adopts the popular encoder–decoder architecture for the semantic segmentation task. The encoding process is used to extract the advanced road semantic features, and the decoding process is used to restore the spatial location information of roads. The encoder not only outputs the high-level semantic feature maps to the decoder, but also outputs the feature maps with detailed spatial information output from encoders at different levels to the decoder at the corresponding level through jump connections. The high-level semantic features captured in the encoding phase play an important role in restoring the spatial details in the decoding stage. Therefore, in order to better restore the road spatial location information and suppress the road occlusion information in the decoder, an encoder with a strong feature learning ability is necessary. As a linear land cover feature, the roads in high-resolution remote sensing images have an obvious long connectivity feature. Therefore, an encoder with strong contextual feature capture capability or global awareness can better extract the features of roads. At present, the mainstream encoders that can better learn the contextual features include the ResNet series, the swin transformer model, and their variants. ResNet is a residual learning framework that is widely used as a feature extractor for semantic segmentation [8], [42]. Multiscale modules and attention mechanisms have also been verified to improve the model's contextual feature learning ability [43], [44]. GAMSNet can significantly improve the road recognition performance by multiscale residual learning and a global awareness operation, and can achieve satisfactory results on simple road situations [25], [45]. Using a pretrained ResNet, swin transformer model, or their variants can achieve outstanding feature extraction results, and can provide sufficient features for the decoder to recover the spatial location information of roads.

Road regions with dense occlusions have a unique texture feature, which is a local smooth texture that is distributed discretely on the strip. In order to achieve explicit modeling of the unique texture feature of road regions with dense occlusions, the spatial strip distribution feature is modeled using strip convolutions, and the local smooth texture is modeled using a constant value, i.e., a hole. The shape of dense

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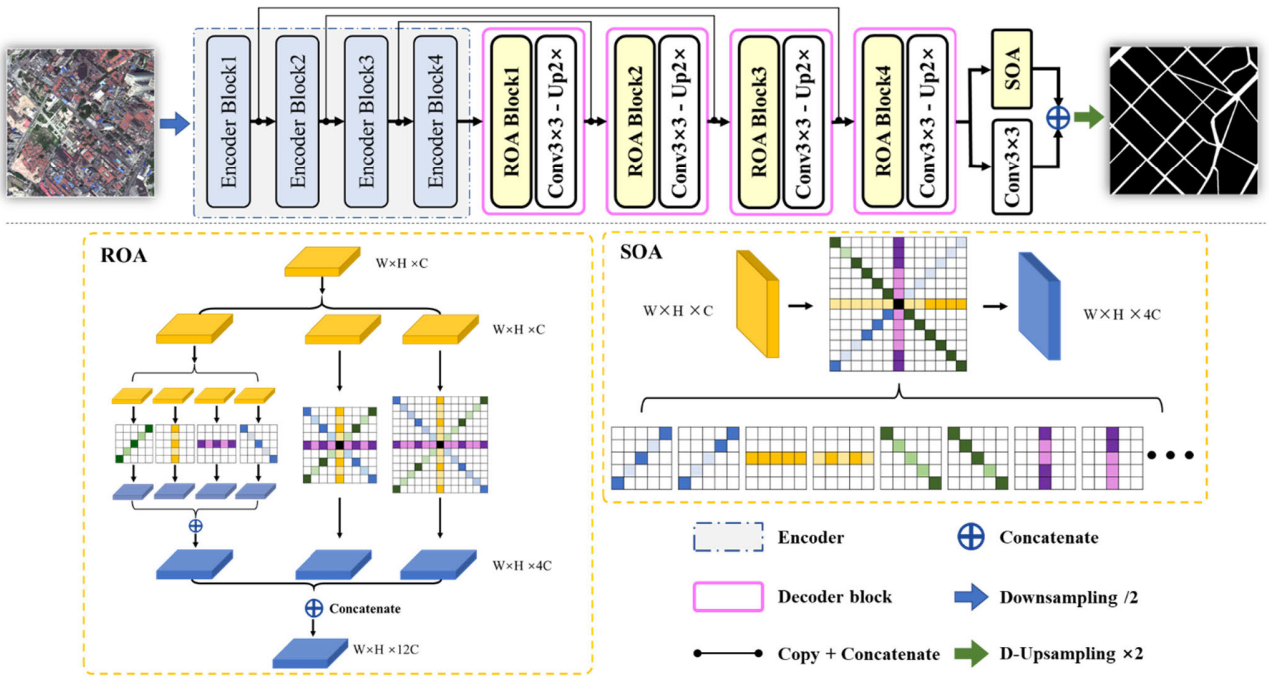


Fig. 2. Overall architecture of the proposed OARENet method. The decoder is the occlusion-aware decoder (OADecoder), which includes four decoder blocks with regular occlusion-aware (ROA) and stochastic occlusion-aware (SOA) modules.

road occlusions is holes in the long and narrow strips with the same material; however, the normal square convolution kernel is not suitable for the feature extraction of linear objects, and simple strip convolutions cannot effectively perceive and handle the information of dense road occlusions. Since the shape feature of a road with occlusions is a strip with holes, strip convolutions with holes are more aligned with the shape of a road with occlusions. Furthermore, strip convolutions with holes can also extract the long connectivity feature of roads without occlusions. At first, the convolution kernel generated by a stochastic sequence is used to achieve explicit modeling of the unique texture, which is the core of the SOA module. However, too many stochastic convolution kernels are difficult to fit, and the spatial details are not abundant in semantic feature maps. Therefore, a regular method based on dilated convolution [46] is used to achieve explicit modeling in the semantic feature maps, to reduce the computational complexity and improve the fit, which is the core of the ROA module. In the OADecoder, there are four decoding blocks, each of which adopts an ROA module. The ROA module is used to identify the road occlusion information before upsampling, and can also extract the shape features of roads with dense occlusions. After this, upsampling is completed in each decoding block, and the feature map is scaled to the original resolution, level by level. By using the features extracted by the occlusion-aware module, not only can the contextual features of the roads be learned to restore the details of the roads, but the expression of road occlusion information can also be suppressed. At the end of the decoder and before the feature result map is output, the SOA module is used to further eliminate the influence of complex road occlusions on the generated road results, and the SOA module is combined as a residual unit for better robustness.

Loss function: There are a large amount of different types of ground objects in high-resolution satellite images. Road pixels typically account for about 10% of the whole image. Because the number of negative samples and positive samples in the road ground-truth labels is very unbalanced, using solely a binary cross-entropy (BCE) loss function can lead to the parameters falling into local optimal solutions, and the model training can be difficult to fit. We introduce dice loss to improve the model training process, which focuses on the mining of the foreground area in the training process. Dice loss performs well for scenes where the positive and negative samples are seriously unbalanced, and is conducive to focusing on the characteristics of roads [47].

B. Occlusion-Aware Decoder

The role of the decoder is to use the high-level semantic features extracted by the encoder to restore the spatial features of the feature map to the original resolution. However, it is difficult for normal convolutions to avoid occlusion information. In order to achieve explicit modeling of the unique texture feature of road regions with dense occlusions, the spatial strip distribution feature is modeled using strip convolutions, and the local smooth texture is modeled using a constant value, i.e., a hole. Strip convolution is totally strip convolution kernel in four different directions. Vertical and horizontal strips can be directly generated by controlling the shape of the convolution kernel. Diagonal strips cannot be directly generated. By offsetting the input feature map by one pixel to the left or right, row by row, the vertical strip achieves oblique sliding relative to the feature map, indirectly achieving a diagonal strip. A stochastic sequence and a regular sequence are used to generate each strip in strip convolution. The strip convolution kernel generated by

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the stochastic sequence, which is the core of the SOA module, fits the detailed texture feature, while the strip convolution kernel generated by the regular sequence, which is the core of the ROA module, reduces the number of parameters and facilitates the model fitting.

ROA module: The ROA module is proposed to enhance the awareness of road regions with dense occlusions. The implementation of the ROA module is divided into three parts: multiscale channels, strip convolution, and regular sequence convolution kernel based on dilated convolution, as shown in Fig. 2. The multiscale channels are implemented by the channel number n and different convolution kernel sizes k. Each channel has four strip convolutions in different directions, and the direction tensor $D_{h,v}$ determines the direction of the convolution kernel $w, h, v \in 0, 1$. A strip convolution kernel with a regular sequence based on dilated convolution is generated by two parameters: the scale value for controlling the multiple scales and the dilation rate. The dilation rate value d decides the holes' regular locations in the convolution kernel.

The ROA module can be described by the calculation process from input to output. As shown in Fig. 2, $X \in R$ denotes the input tensor for the ROA module. In the ROA module, the input tensor X is first copied n times for the different-scale channels (set to 3 by default), which are then fed into the designed convolutional blocks with different scales k. In each convolutional block, to implement strip convolution with holes in different directions, the input tensor X is copied four times onto the convolution kernel w. Each copied X is calculated with a strip convolution kernel. The output feature map $Z^X_{D[h,v]}$ of the four different-direction channels is then concatenated. The output feature map of the different-scale channels is concatenated to obtain the output of the occlusion-aware module $F(X)$. The ROA module can be formulated as follows:

$$w_d = A^d_{1*2k} \tag{1}$$

$$Z = X * w_d = \sum_{l=-k}^{k} X\left[i + D_h l,\, j + D_v l\right] \cdot w_d[l] \tag{2}$$

$$F(X) = \sum_{m=1}^{n} G_m\Big(\sum_{h=0}^{1}\sum_{v=0}^{1} Z^X_{D[h,v]}\Big) \tag{3}$$

where i and j represent the index of the input tensor X, l represents the index of the strip convolution kernel w, G represents the function of the channel, and A represents the regular sequence based on dilated convolution.

SOA module: After four OADecoder blocks, the resolution of the feature map is restored to half that of the input image. At this point, the spatial details of the feature map are very rich, so that the scale and geometric characteristics of the roads and the types and shapes of the occlusions in the roads are more complex. In order to enable the decoder to adaptively learn more types of road occlusion spatial detail information, to output better road results, we propose the SOA module, which is generated by a stochastic sequence, so that the convolution kernel can match more road and occlusion situations, like a template, and extract regions with a high probability of road occlusions. As shown in Fig. 2, the customized stochastic strip convolution captures road features of more types and scales, and at the same time, the types of perceived occlusions also increase significantly. Therefore, SOA convolution can better reduce the impact of occlusions on the road, and can improve the ability of the model to capture road features of different scales. As shown in Fig. 2, different types of convolution kernels are generated, which play different roles in capturing the road occlusion shape feature. In the example, the purple convolution core in the strip convolution can better capture the features of short-distance or zigzagging roads, while the other three strip convolutions can perceive the occlusions under different road conditions, thus improving the model's perception of roads and road occlusions.

The difference between the implementation process of the SOA and ROA modules is the generation of the strip convolution. The SOA module also has multiscale channels and different-direction strip convolution kernels. The strip convolution kernel of the SOA module is generated by a stochastic sequence. The stochastic sequence is defined in PyTorch as $\boldsymbol{w_s} = \boldsymbol{R_{1*s}}$ by a random matrix, where s represents the length of the strip convolution kernel and R represents the random matrix function, which controls the distribution position representing local smooth texture constants. By combining different numbers of stochastic sequence convolution kernels, the SOA module can fit complex road regions with dense occlusions.

IV. Experiments and Results

In this section, in order to verify the effectiveness of the proposed OADecoder, we describe the experiments conducted based on the DeepGlobe road dataset and the newly built JHWV dense occlusion road dataset. We also describe the experiments conducted with large-scale images of Wuhan and Shanghai, to verify the generalization ability of the proposed method. In order to verify the function of the proposed OARENet method, mainstream encoders were used as the feature extractors to verify the effect of the OADecoder. The comparison methods were LinkNet50 [38], LinkNet50 + OADecoder, GAMSNet [25], GAMSNet + OADecoder, swin transformer, and swin transformer + OADecoder. The LinkNet50, GAMSNet, and swin transformer methods respectively used ResNet50, GAMSNet, and swin transformer as the encoders to extract the road features, and used the decoder of LinkNet50 as the decoder. The LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder methods respectively used the above encoders to extract the road features, and used the proposed OADecoder as the decoder.

A. Experimental Settings

Implementation details: All the experiments were conducted in PyTorch [48]. The training set was randomly cut to 512 × 512 pixels in size, and horizontal, vertical, and diagonal flipping data enhancement methods were used. The loss function was the sum of BCE loss and dice loss. The training optimizer for all the models was the Adam optimizer [49]. The models were trained for 150 epochs, the batch size was eight images,

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and the initial learning rate was 2e-4, which was divided by 5 at epochs {90, 110, and 130}. Finally, the output probability map of the model used 0.5 as the threshold to generate the final road segmentation map. The experiments were run on two NVIDIA Tesla P100 GPUs (with 16-GB GPU memory).

Evaluation metrics: The metrics of precision, recall, F1-score, and intersection over union (IoU) are typically used to evaluate the quality of semantic segmentation model prediction results. The above indicators can be calculated using a confusion matrix. However, the precision and recall are a pair of mutually restrictive measures. Generally speaking, only in simple tasks can both indicators be high. Therefore, when considering the model capability in a comprehensive manner, the harmonic average value F1-score of the accuracy and recall can be used as an evaluation metric. The IoU is the ratio between the intersection of predicted road pixels and real road pixels and their union. Among the evaluation metrics, the F1-score and IoU are the more comprehensive metrics. The metrics are defined as shown in the following equations:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{4}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{5}$$

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{6}$$

$$\mathrm{IOU} = \frac{P \times R}{R - P \times R + P} = \frac{\mathrm{TP}}{\mathrm{FN} + \mathrm{TP} + \mathrm{FP}} \tag{7}$$

where TP represents the number of true positives, TN represents the number of true negatives, FP represents the number of false positives, and FN represents the number of false negatives.

B. Experiment 1: DeepGlobe Dataset

The DeepGlobe dataset was published as part of the CVPR DeepGlobe Road Extraction Challenge in 2018 [7]. The dataset contains 8570 satellite images from India, Indonesia, and Thailand, each with a size of 1024 × 1024 pixels and a spatial resolution of 0.5 m. This dataset covers the land cover of various scenes, including both rural and urban areas. It should be noted that, among the 8570 satellite images in the DeepGlobe dataset, 6226 images with labels are provided as the training set, and 2344 images are provided as the verification set and test set, for which no corresponding image labels have been published. Therefore, the 6226 images with labels were divided into 4696 sample pairs as the training set and 1530 sample pairs were used as the test set in this experiment.

We compared the performance of the proposed OADecoder with that of several state-of-the-art methods for road extraction on the DeepGlobe road dataset, i.e., DeepRoadMapper [32], Topology loss [50], RoadCNN [34], ResUNet [21], LinkNet50 [38], the compact generalized nonlocal (CGNL) network [51], the global context network (GCNet) [52], the convolutional block attention module (CBAM) [53], the globally aware road detection network with multiscale residual learning (GAMSNet) [25], the connectivity attention module network (CoANet) [42], and swin transformer [45]. The DeepRoadMapper, RoadCNN, and Topology loss methods generate vector roads by segmentation and postprocessing. The ResUNet and LinkNet50 methods are baseline models for the sematic segmentation task. The CGNL, GCNet, and CBAM methods can capture more adequate contextual features in the residual blocks. GAMSNet and CoANet are state-of-the-art methods for road extraction, which can learn the long connectivity feature of roads.

Fig. 3 shows the visualization results of road extraction by the different methods on the DeepGlobe dataset. The odd rows show the prediction results for the two DeepGlobe images. The even rows show enlarged views of the local areas corresponding to the odd rows above. Most images in the DeepGlobe dataset belong to rural areas, where roads are easily occluded by roadside trees. The DG-1 and DG-2 images show that the rural road occlusions are caused by roadside trees. In these cases, the extracted results of LinkNet50, GAMSNet, and swin transformer show that the roads with tree occlusions are fragmented. However, the extracted results of the three models (LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder) which use the OADecoder as the decoder are not affected by the tree occlusions and the extracted roads are connected. Although the dense occlusions destroy the connectivity and shape feature of the road, the road region still has a local smooth texture from the road surface. The proposed method can extract road regions using the local smooth texture. This shows that the proposed method can weaken the impact of tree occlusions on rural areas. The visualized results verify the superiority of the proposed OADecoder in handling roads with occlusions. The OADecoder weakens the impact of occlusions and achieves a better performance in both urban and rural areas. The visualized results also verify that the OADecoder can refine the extracted roads with different encoders.

Table I lists the quantitative results of the different methods. The proposed LinkNet50 + OADecoder method achieves an IoU of 67.53%, which is better than LinkNet50 but less than that of the swin transformer. The GAMSNet + OADecoder and swin transformer + OADecoder methods achieve IoU scores of 69.28% and 69.25%, which are better results than those of the other methods. Furthermore, the GAMSNet + OADecoder and swin transformer + OADecoder methods show an improvement over the original GAMSNet and swin transformer methods of 0.44% and 0.7%, respectively. However, it should be noted that the satellite imagery used in the DeepGlobe dataset was mostly sampled from rural areas of Southeast Asia, where dense occlusions are less common.

C. Experiment 2: JHWV Dense Occlusion Road Dataset

In order to further evaluate the ability of the proposed method to deal with road occlusions, we built a new dense occlusion road dataset: the JHWV dataset. The image source was WorldView-2, and the covered area is Jianghan District, Wuhan, China. We produced a finely annotated dataset of roads in urban areas with dense buildings, including 867 images and the corresponding road labels. The size of each image is 1024 × 1024 pixels, and the image spatial resolution

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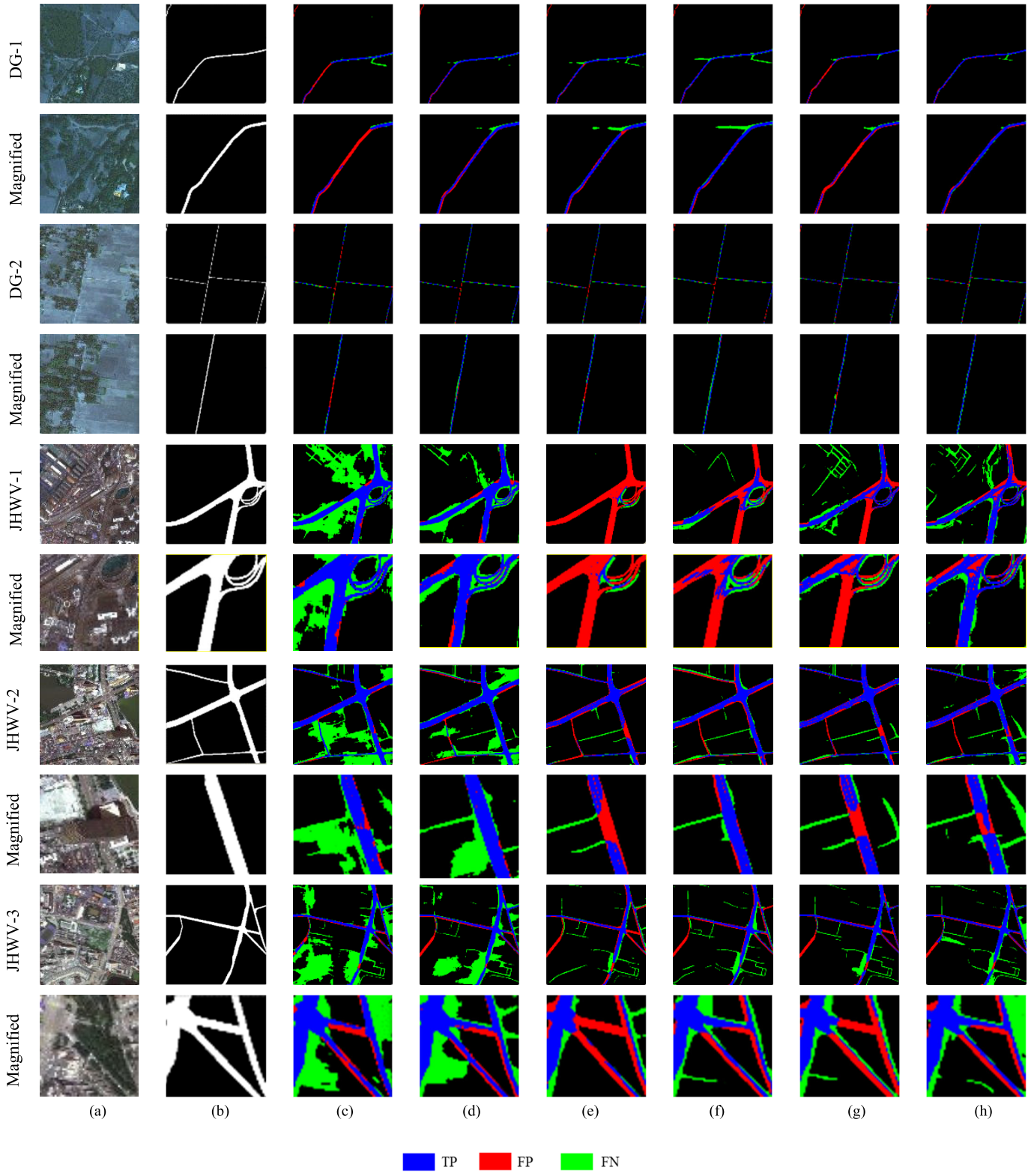


Fig. 3. Visual comparison of the different methods on the DeepGlobe dataset and the JHWV dense occlusion road dataset. (a) Image. (b) Road map. (c) LinkNet50. (d) LinkNet50 + OADecoder. (e) GAMSNet. (f) GAMSNet + OADecoder. (g) SwinT. (h) SwinT + OADecoder.

is 0.5 m. The characteristic of this dataset is that it only includes urban areas, so that there are dense urban buildings in the images, with various road types, complex shapes, and many roads surrounded by buildings and trees. The images are manually labeled at the pixel level, which is more accurate than the labels generated by OpenStreetMap (OSM). This dataset can effectively verify the road recognition ability of a model in urban areas, and can make up for the shortcomings of the current road extraction models with remote sensing images of urban roads with dense buildings.

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TABLE I
QUANTITATIVE COMPARISON OF THE PROPOSED OADECODER WITH THE STATE-OF-THE-ART METHODS ON THE DEEPGLOBE DATASET (%)

Methods	P	R	F1	IoU
DeepRoadMapper [32]				62.58
Topology loss (with BCE) [50]				64.95
Topology loss (with SoftIoU) [50]				64.94
RoadCNN [34]				67.61
Improved connectivity [54]				67.21
ResUNet [21]	79.16	78.55	78.63	65.04
LinkNet50 [38]	81.79	78.60	80.16	66.89
CGNL[51]	82.04	78.84	80.41	67.23
GCNet [52]	82.33	79.60	80.94	67.98
CBAM [53]	81.56	80.52	81.04	68.12
GAMSNet [25]	83.14	80.00	81.54	68.84
CoANet(2022) [42]	74.61	86.38	80.06	66.75
Swin transformer [45]	80.47	81.67	81.06	68.14
LinkNet50 + OADecoder (proposed)	80.74	80.77	80.76	67.53
GAMSNet + OADecoder (proposed)	80.65	83.09	**81.85**	**69.28**
SwinT + OADecoder (proposed)	81.72	81.95	**81.84**	**69.25**

Fig. 3 shows the visualization results of road extraction by the different methods on the JHWV dense occlusion road dataset. The three images selected in the JHWV dataset belong to urban areas, and the dense road occlusions are caused by buildings or building shadows. In the JHWV-1 image, there are three types of classical dense occlusion situations, i.e., high-rise buildings, shadows, and street trees. In this image, although LinkNet50 can detect most of the roads with shadow occlusions, it also extracts large areas of buildings as roads, which leads to the high recall score and poor precision score, while LinkNet50 + OADecoder can greatly reduce the high recall score and generate better road extraction results. GAMSNet + OADecoder and swin transformer + OADecoder, which use the OADecoder, significantly improve the extracted roads with road shadow occlusions, and the results are better than the extracted results of GAMSNet and swin transformer. The magnified JHWV-1 image represents dense occlusions caused by shadows, where most of the road region is covered by shadows, resulting in missing roads in the extracted result of the baseline. The magnified JHWV-2 image represents dense occlusions by high-rise buildings, which affect the connectivity of the extracted road region. The magnified JHWV-3 image represents dense occlusions caused by street trees, where only a small portion of the road surface is visible, which also results in missing roads in the extracted results of the baseline. In the JHWV-1 image, only the swin transformer + OADecoder method can extract complete road regions, the reason for which is that the high-rise buildings affect the texture of the road region, but there is still local texture that is normal. The occlusion-aware module relies solely on local texture to recognize road regions with dense occlusions. In the JHWV-2 and JHWV-3 images, the extracted results of LinkNet50, GAMSNet, and swin transformer are all affected by the occlusions, and the extracted roads are fragmented. However, the extracted roads of the LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder models are connected and are not affected. For the dense occlusions of the JHWV-2 image by high-rise buildings, the stochastic sequence kernel of the SOA module can fit this special occlusion situation, based on the feature of the roads on both sides of the building. The dense occlusion of the JHWV-3 image by continuous street trees destroys the road shape and connectivity feature, but a portion of the smooth road surface in the middle road region is still displayed in the image, which is helpful for the OADecoder to extract the road. These results show that the proposed method can indeed weaken the impact of dense road occlusions on urban areas.

It can be seen from Table II that the performance of GAMSNet is worse than that of LinkNet50, which is consistent with the problem of the dense occlusions destroying the long connectivity feature, and it is difficult for GAMSNet to capture the long connectivity feature of roads to extract roads with dense occlusions. Compared with the baseline models of LinkNet50, GAMSNet, and swin transformer, the three models of LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder using the OADecoder improve the IoU score by 5.78%, 8.55%, and

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TABLE II
QUANTITATIVE COMPARISON OF THE PROPOSED OADECODER WITH THE MAINSTREAM ENCODERS ON THE JHWV DATASET (%)

	Precision	Recall	F1	IoU
LinkNet50	37.86	70.18	49.19	32.39
LinkNet50 +OADecoder	47.51	66.01	**55.25**	**38.17**
GAMSNet	71.81	35.81	47.79	31.26
GAMSNet +OADecoder	74.05	44.04	**55.23**	**38.15**
Swin transformer	64.01	50.70	56.58	39.46
Swin transformer +OADecoder	66.05	53.79	**59.29**	**42.14**

2.68% respectively, which shows that the OADecoder can be applied to different encoders, especially when the encoder has the ability to capture contextual features. The OADecoder can play an important role in effectively suppressing occlusion information in the process of restoring the spatial details, and can significantly improve the performance of the model in extracting roads with dense occlusions.

D. Experiment 3: Large-Scale Images

This dataset consists of two large-scale high-resolution images from Wuhan (WH) and Shanghai (SH) in China [25], covering 203.63 km^2 in total. The sizes of the WH and SH images are 21 624 × 24 196 and 16 768 × 16 640 pixels, respectively, and the image spatial resolution is 0.5 m. The images are representative high-resolution remote sensing images for practical application, which cover urban areas and contain many dense high-rise buildings surrounding the roads.

Dense road occlusions in large-scale high-resolution remote sensing images exist most frequently in urban areas, and pose a great challenge to the robustness of road extraction models. In order to verify the generalization ability of the proposed method to solve the dense road occlusion problem, this experiment was carried out on the large-scale images. The model was trained on the DeepGlobe training set and directly tested on the large-scale images. The comparison methods were LinkNet50, LinkNet50 + OADecoder, GAMSNet, GAMSNet + OADecoder, swin transformer, and swin transformer + OADecoder.

Fig. 4 shows the visualization results of the proposed method of swin transformer + OADecoder on the large-scale Wuhan image. For a detailed visual comparison, we selected three local areas (Wuhan-1, Wuhan-2, and Wuhan-3) to compare the details, as shown in Fig. 5. As shown in the Wuhan-1 and Wuhan-2 subimages, the main road occlusions in the imagery are caused by the high-rise buildings beside the roads. In these cases, the extracted roads of LinkNet50, GAMSNet, and swin transformer are fragmented, but GAMSNet is the least affected. After adding the proposed OADecoder, the extracted results are significantly improved. In the Wuhan-3 subimage, because the building and tree occlusions are very serious, the main roads are missing in the extracted results of LinkNet50, GAMSNet, and swin transformer. However, LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder can completely extract the main roads with occlusions, and the extracted results are significantly improved. This proves that the OADecoder can improve the ability of road extraction for different encoders. When dense road occlusions exist in the imagery, the improvement performance is obvious. The corresponding quantitative results are listed in Table III. Among the baseline models of LinkNet50, GAMSNet, and swin transformer, the swin transformer model shows the best performance, 2.04% and 1.74% higher than the other models, respectively. After the

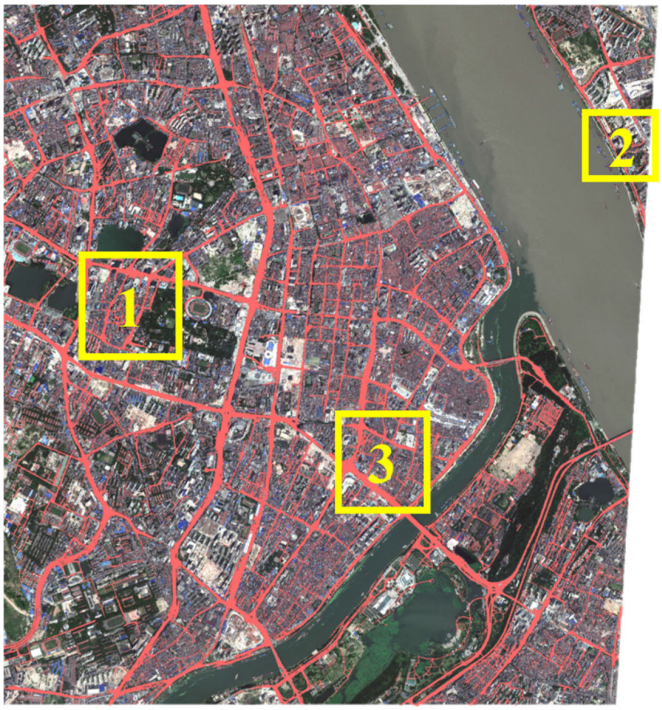


Fig. 4. Visual result of swin transformer + OADecoder on the large-scale Wuhan image. Red: predicted roads.

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TABLE III
QUANTITATIVE COMPARISON OF THE PROPOSED OADECODER WITH THE MAINSTREAM ENCODERS ON THE LARGE-SCALE WUHAN IMAGE (%)

Method	P	R	F1	IoU
LinkNet50	68.26	43.21	53.01	37.07
LinkNet50 +OADecoder	80.58	46.49	**59.12**	**41.97**
GAMSNet	75.00	42.69	54.41	37.37
GAMSNet +OADecoder	74.34	47.99	**58.32**	**41.17**
Swin transformer	75.02	44.96	56.23	39.11
Swin transformer +OADecoder	73.92	48.20	**58.35**	**41.19**

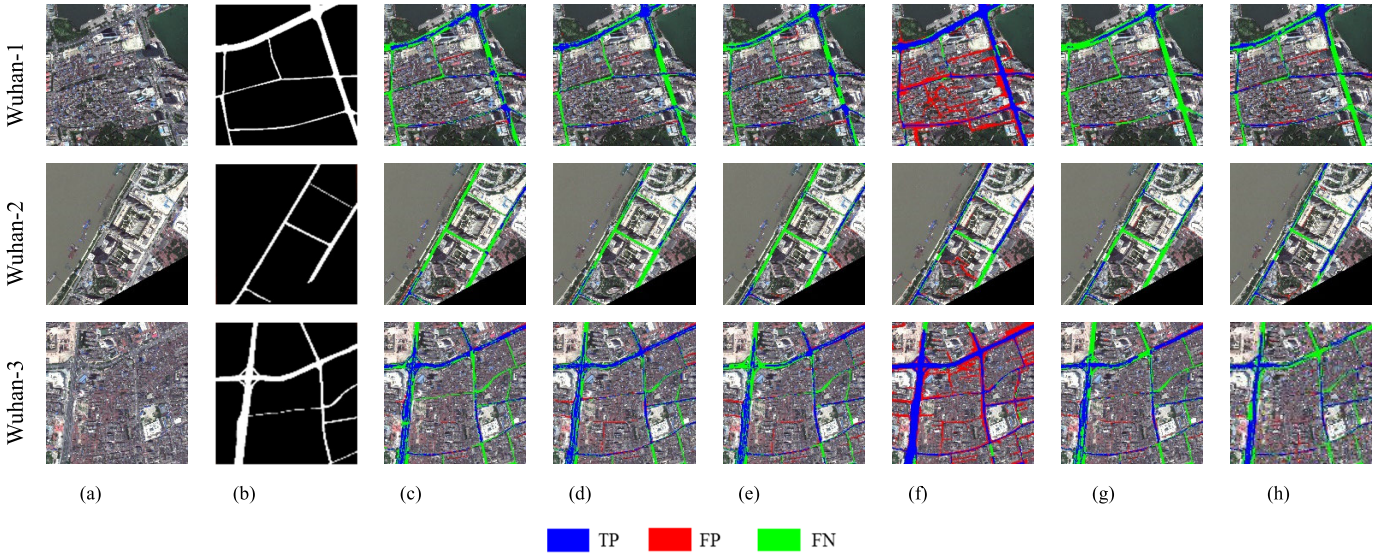


Fig. 5. Visual comparison of the different methods on three local areas in the large-scale Wuhan image. (a) Image. (b) Road map. (c) LinkNet50. (d) LinkNet50 + OADecoder. (e) GAMSNet. (f) GAMSNet + OADecoder. (g) SwinT. (h) SwinT + OADecoder.

TABLE IV
QUANTITATIVE COMPARISON OF THE PROPOSED OADECODER WITH THE MAINSTREAM ENCODERS ON THE LARGE-SCALE SHANGHAI IMAGE (%)

Method	P	R	F1	IoU
LinkNet50	91.99	32.44	47.97	31.55
LinkNet50 + OADecoder	91.73	38.11	**53.85**	**36.84**
GAMSNet [25]	87.70	48.31	62.30	45.25
GAMSNet + OADecoder	72.87	57.27	**64.20**	**47.27**
Swin transformer	88.18	45.97	60.44	43.31
Swin transformer + OADecoder	83.37	48.54	**61.35**	**44.25**

OADecoder is added to the corresponding model, the IoU scores of LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder are increased by 4.90%, 3.80%, and 2.08%, respectively. The swin transformer +

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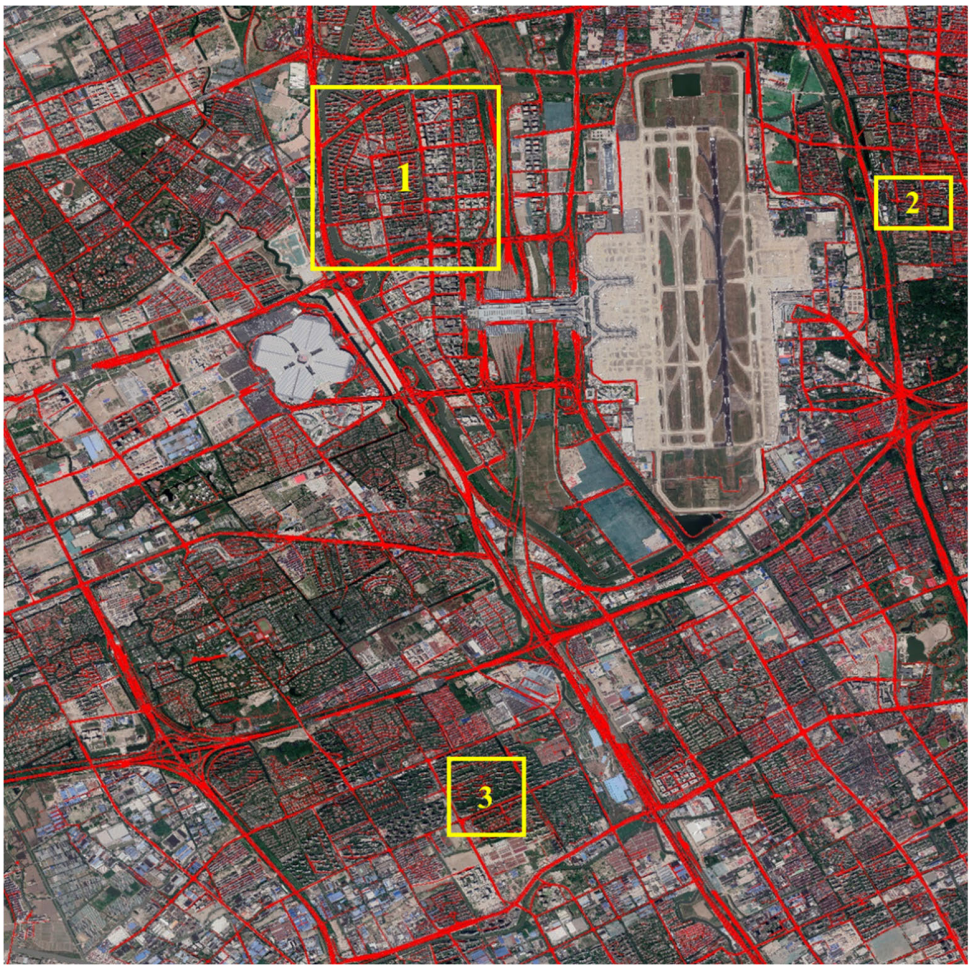


Fig. 6. Visual result of swin transformer + OADecoder on the large-scale Shanghai image. Red: predicted road networks.

OADecoder model shows the best road extraction performance, which demonstrates the robustness and superiority of the proposed method.

Fig. 6 shows the visualization results of swin transformer + OADecoder on the Shanghai image. In the same way, for a detailed visual comparison, we selected three local areas to compare the details. As shown in Fig. 7, the road occlusions are caused by the buildings and trees in the urban residential areas. LinkNet shows a serious missing road problem in the urban residential areas, and the LinkNet + OADecoder model does not show a great improvement, which is because the ResNet encoder has the weakest ability to capture contextual features. The extracted result of the GAMSNet model is better than that of LinkNet. However, when the road is mostly blocked by buildings, the whole road is missed. GAMSNet + OADecoder can effectively improve the extracted result of GAMSNet, but there is still a large number of missing roads with occlusions in residential areas. The result of swin transformer is slightly better than that of GAMSNet, but the extracted result of swin transformer + OADecoder is very good. The swin transformer + OADecoder model can not only completely extract the main roads, but the extracted result for the small roads in the residential areas is better than that of the other models. This proves that the proposed occlusion-aware module can effectively solve the problem of dense road occlusions, and the model based on swin transformer as the encoder shows the best performance in the roads with dense occlusions. The corresponding quantitative results are listed in Table IV. Among the baseline models of LinkNet50, GAMSNet, and swin transformer, GAMSNet shows the best performance, which is 13.7% and 1.94% higher than that of the other models, respectively. After the OADecoder is added to the corresponding model, the IoU scores of LinkNet50 + OADecoder, GAMSNet + OADecoder, and swin transformer + OADecoder are increased by 5.29%, 2.02%, and 0.94%, respectively. These results prove that the OADecoder can effectively improve the ability to extract roads with dense occlusions.

V. Analysis of the Occlusion-Aware Decoder

A. Heatmap Analysis

In the above experiments, we found that the proposed OADecoder can improve the ability of road extraction with dense occlusions. The reason for this is that the OADecoder can perceive and suppress the road occlusion information, allowing the model to more easily extract roads with occlusions. To verify the effect of the proposed OADecoder, the heatmaps for the OADecoder are presented to visually analyze the impact of the OADecoder on the occlusion information awareness.

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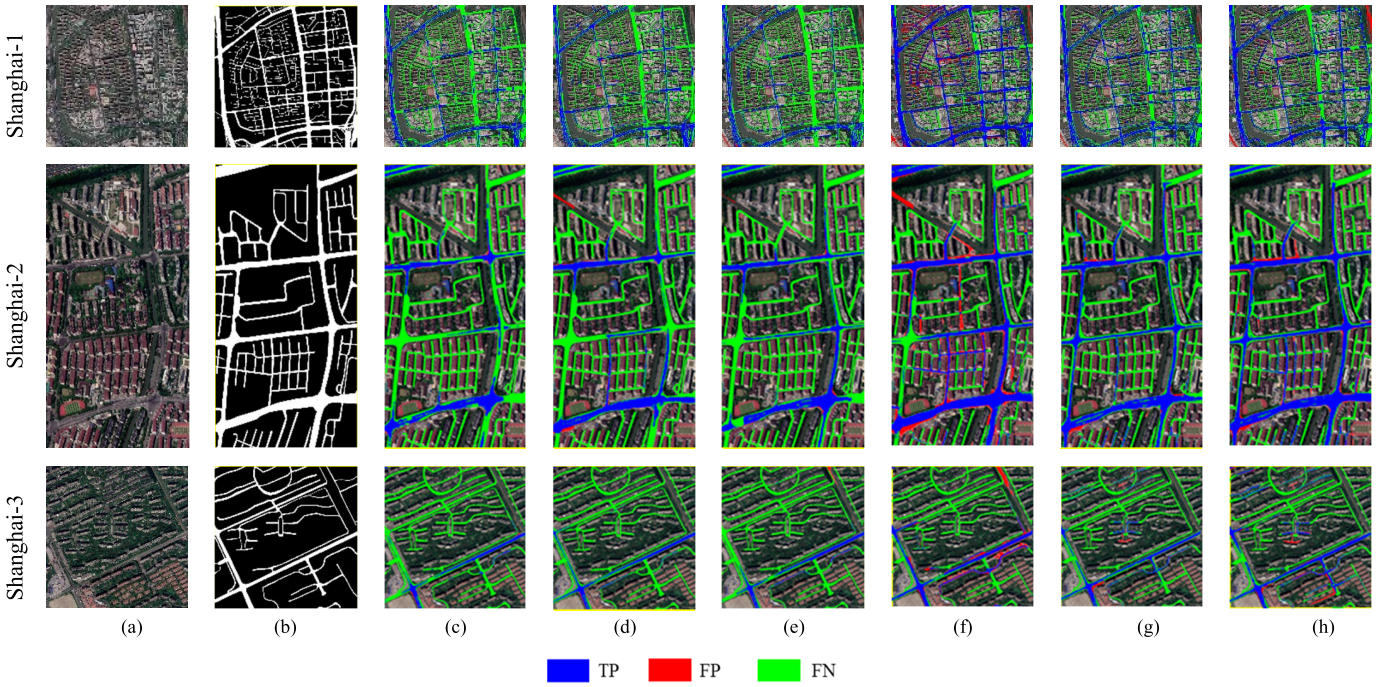


Fig. 7. Visual comparison of the different methods on three local areas in the Shanghai image. (a) Image. (b) Road map. (c) LinkNet50. (d) LinkNet50 + OADecoder. (e) GAMSNet. (f) GAMSNet + OADecoder. (g) SwinT. (h) SwinT + OADecoder.

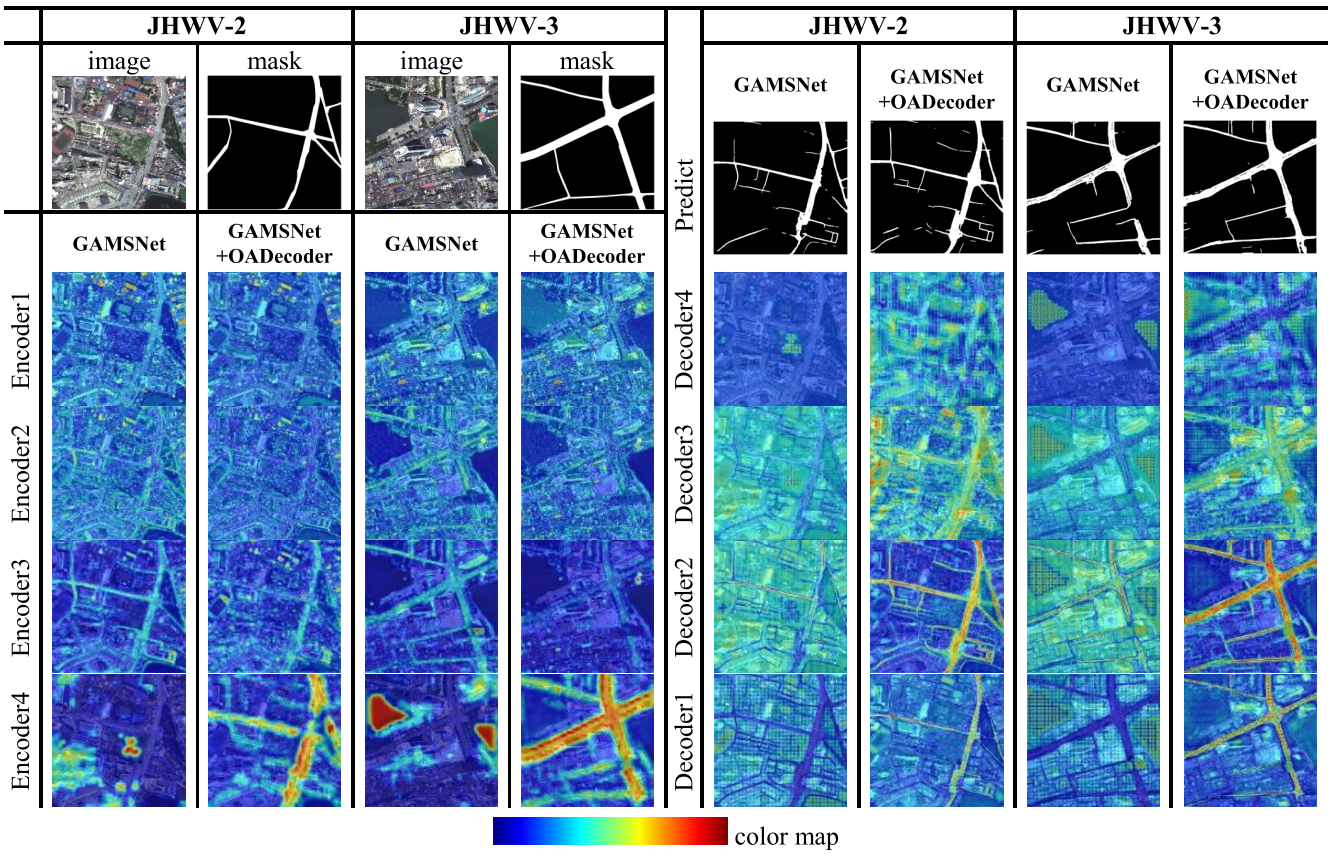


Fig. 8. Heatmaps of GAMSNet and GAMSNet + OADecoder on the JHWV dataset.

The heatmaps of the weights in the different decoder stages for the two images are shown in Fig. 8. To visualize the impact of the modules on the whole model, we also mapped the heatmaps of the different encoder stages.

From Fig. 8, it can be seen that, at the stage of Encoder1 and Encoder2, the weights of GAMSNet and GAMSNet + OADecoder are basically the same, and neither of the models pay attention to the road. The reason for this is that the shallow

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TABLE V
ABLATION STUDY (%) FOR THE PROPOSED ROA AND SOA MODULES ON THE DEEPGLOBE AND JHWV DATASETS. BASELINE IS GAMSNET (NO. 1)

			DeepGlobe		JHWV	
No.	ROA	SOA	F1	IoU	F1	IoU
1			81.54	68.84	47.79	31.26
2		√	81.57	68.88	51.80	34.95
3	√		**82.17**	**69.73**	50.27	33.58
4	√	√	81.85	69.28	**55.23**	**38.15**

features have no ability to capture road features. At the stage of Encoder3, the weights of the roads can be distinguished. However, both models still do not pay attention to the road areas with occlusions. At the stage of Encoder4, it can be clearly seen that the weight of GAMSNet + OADecoder results in it paying more attention to the roads and road occlusions, outperforming the original GAMSNet model. This proves that the OADecoder helps the encoder to focus on the roads with occlusions, which are more difficult to extract. At the stage of Decoder4, the heatmap is obtained by interpolation of the weight map with a small size, and does not provide visual information. At the stage of Decoder3, compared with GAMSNet, GAMSNet + OADecoder can focus more on the roads with road occlusions. At the stage of Decoder2 and Decoder1, the road network is obvious, and although the OADecoder successfully generates roads with occlusions, the occlusions still affect the weights in the heatmaps. The above findings from the heatmaps indicate that the OADecoder can help the model to perceive and suppress the road occlusion information, and reduce the impact of occlusions on the spatial information recovery process.

B. Ablation Study

In the OADecoder, the ROA and SOA modules are proposed to perceive and suppress road occlusion information and explore the spatial detail of different occlusions. To validate the effectiveness of the two modules, we conducted experiments with different configurations on the JHWV dataset and DeepGlobe dataset, as shown in Table V. The baseline (No. 1) is the original GAMSNet model, which can capture the long connectivity feature of roads. After adding the proposed ROA and SOA modules to the baseline, the performance in terms of the IoU score increases from 31.26% to 34.95% and 33.58% on the JHWV dataset, and the performance in terms of the IoU score increases from 68.84% to 69.73% and 68.88% on the DeepGlobe dataset. After adding both the ROA and SOA modules to the OADecoder, 6.89% and 0.44% improvements in the IoU score are seen on the JHWV dataset and DeepGlobe dataset, respectively. From the quantitative experimental results of JHWV, it is clear that the SOA module alone has more gain compared to the ROA module alone when facing the complex problem of road occlusion in the city. However, from the DeepGlobe dataset, the ROA module

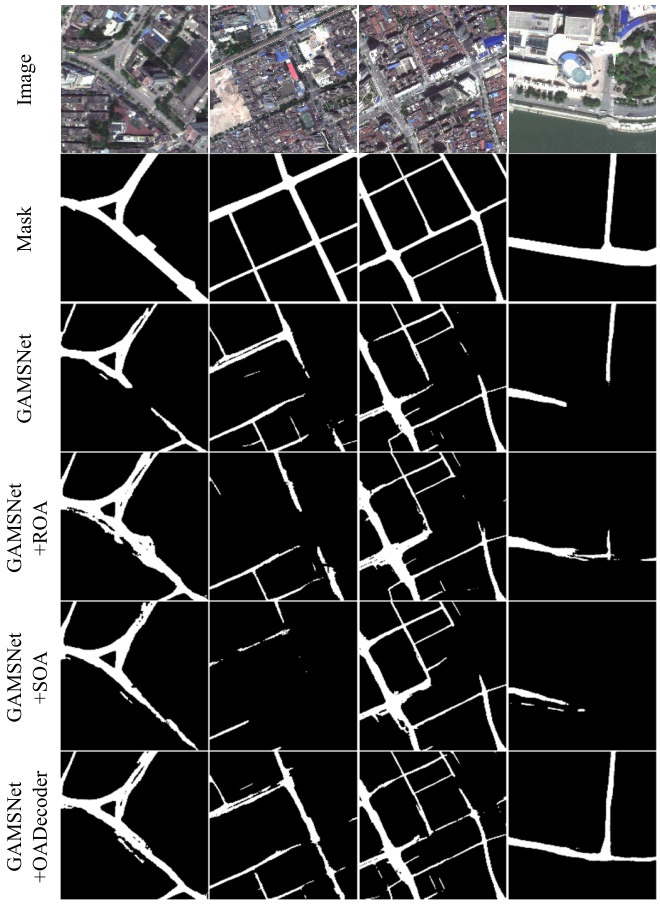


Fig. 9. Visual comparison of the proposed ROA and SOA on the JHWV dataset.

achieves better gain, and we hypothesize that the reason is due to the fact that the robustness of the ROA module is more important for simple occlusion situation or unobscured road situation, but the SOA module is more suitable for complex road occlusion problems. As shown in Fig. 9, both ROA and SOA improve compared to the original GAMSNet in facing the occlusion problem caused by buildings and trees, but only OADecoder performs better compared to both ROA and SOA alone. These results verify that the two proposed modules are effective for road extraction, and can achieve better results when combined. It should be noted that most images in the DeepGlobe dataset are from rural areas, which do not have complex dense road occlusions. Therefore, the SOA module is just applicable to roads with dense occlusions, and the ROA module is helpful for different road situations.

C. Model Complexity and Time Complexity Analysis

In order to comprehensively evaluate the value of the model, we adopted the number of parameters (# params) to measure the model complexity and the giga floating-point operations per second (GFLOPs) to measure the theoretical computational overhead. We have compared the model complexity and time complexity of the proposed method and the other models. The GFLOPs were calculated under the same conditions and the input images were all set to one batch. The parameter

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TABLE VI
MODEL COMPLEXITY AND TIME COMPLEXITY OF THE PROPOSED METHOD AND SOME OF THE STATE-OF-THE-ART METHODS

	Params(M)	GFLOPs
LinkNet	28.83	56.98
DLinkNet	217.79	121.21
GAMSNet	29.33	58.32
SwinTransformer	28.27	36.43
CoANet	59.04	276.70
SwinTransformer +OADecoder	33.48	36.27

settings of the proposed method were consistent with the default settings in the manuscript. Compared to the other road extraction methods, the model parameters and time complexity of the proposed method are satisfactory. The OADecoder that we proposed does not significantly increase the model parameters and time complexity, as shown in Table VI.

VI. CONCLUSION

In this article, the OARENet has been proposed to solve the dense road occlusion problem for high-resolution remote sensing imagery. In OARENet, the OADecoder is proposed to suppress occlusion information, which can effectively improve the ability of different encoders to recognize roads and improve the ability of the models to extract roads with dense occlusions. Our proposed OARENet is effective for the occlusion problem in urban areas without affecting the extraction accuracy of conventional roads. Our method has a wide range of application scenarios in the task of road extraction in urban areas, and since the open-source community OSM has provided a large number of labeled samples of urban areas, our method can be directly applied to any city in the world through fine-tuning by utilizing the existing labeled samples. Incorporating the unique shape information of the road in the occluded area helps in the extraction of the occluded road area, and designing a better feature learning method for the target-specific features can provide better results for the specific target. However, the disconnection caused by the occlusion problem is not obvious in the pixel-level results, but it will have a great impact on the connectivity of the road vectors, but it is still difficult to apply the latest works in road vectorization [34], [55], [56], so we will challenge the road vectorization task in the future work.

REFERENCES

[1] J. Cui, F. Liu, D. Janssens, S. An, G. Wets, and M. Cools, "Detecting urban road network accessibility problems using taxi GPS data," *J. Transp. Geography*, vol. 51, pp. 147–157, Feb. 2016.

[2] Z. Zhong, J. Li, W. Cui, and H. Jiang, "Fully convolutional networks for building and road extraction: Preliminary results," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 1591–1594.

[3] D. Frey, M. Butenuth, and D. Straub, "Probabilistic graphical models for flood state detection of roads combining imagery and DEM," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 6, pp. 1051–1055, Nov. 2012.

[4] J. B. Mena, "State of the art on automatic road extraction for GIS update: A novel classification," *Pattern Recognit. Lett.*, vol. 24, no. 16, pp. 3037–3058, Dec. 2003.

[5] Q. Li, L. Chen, M. Li, S.-L. Shaw, and A. Nuchter, "A sensor-fusion drivable-region and lane-detection system for autonomous vehicle navigation in challenging road scenarios," *IEEE Trans. Veh. Technol.*, vol. 63, no. 2, pp. 540–555, Feb. 2014.

[6] C. Wen, X. Sun, J. Li, C. Wang, Y. Guo, and A. Habib, "A deep learning framework for road marking extraction, classification and completion from mobile laser scanning point clouds," *ISPRS J. Photogramm. Remote Sens.*, vol. 147, pp. 178–192, Jan. 2019.

[7] I. Demir et al., "DeepGlobe 2018: A challenge to parse the Earth through satellite images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 172–17209.

[8] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: LinkNet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," presented at the *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 192–1924.

[9] F. Schug, A. Okujeni, J. Hauer, P. Hostert, J. Ø. Nielsen, and S. van der Linden, "Mapping patterns of urban development in Ouagadougou, Burkina Faso, using machine learning regression modeling with bi-seasonal Landsat time series," *Remote Sens. Environ.*, vol. 210, pp. 217–228, Jun. 2018.

[10] J. Susaki, M. Kajimoto, and M. Kishimoto, "Urban density mapping of global megacities from polarimetric SAR images," *Remote Sens. Environ.*, vol. 155, pp. 334–348, Dec. 2014.

[11] Y. Chen, Y. Li, H. Zhang, L. Tong, Y. Cao, and Z. Xue, "Automatic power line extraction from high resolution remote sensing imagery based on an improved radon transform," *Pattern Recognit.*, vol. 49, pp. 174–186, Jan. 2016.

[12] S. Das, T. T. Mirnalinee, and K. Varghese, "Use of salient features for the design of a multistage framework to extract roads from high-resolution multispectral satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3906–3931, Oct. 2011.

[13] Z. Sun, H. Fang, M. Deng, A. Chen, P. Yue, and L. Di, "Regular shape similarity index: A novel index for accurate extraction of regular objects from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3737–3748, Jul. 2015.

[14] M. Barzohar and D. B. Cooper, "Automatic finding of main roads in aerial images by using geometric-stochastic models and estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 7, pp. 707–721, Jul. 1996.

[15] D. Chai, W. Förstner, and F. Lafarge, "Recovering line-networks in images by junction-point processes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 1894–1901.

[16] I. Laptev, H. Mayer, T. Lindeberg, W. Eckstein, C. Steger, and A. Baumgartner, "Automatic extraction of roads from aerial images based on scale space and snakes," *Mach. Vis. Appl.*, vol. 12, no. 1, pp. 23–31, Jul. 2000.

[17] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[18] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[19] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, 2015, pp. 234–241.

[20] G. Cheng, Y. Wang, S. Xu, H. Wang, S. Xiang, and C. Pan, "Automatic road detection and centerline extraction via cascaded end-to-end convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3322–3337, Jun. 2017.

[21] Z. Zhang, Q. Liu, and Y. Wang, "Road extraction by deep residual U-Net," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 5, pp. 749–753, May 2018.

[22] V. Mnih, "Machine learning for aerial image labeling," Ph.D. dissertation, Univ. Toronto, Toronto, ON, Canada, 2013. [Online]. Available: https://www.cs.toronto.edu/ vmnih/docs/Mnih_Volodymyr_PhD_Thesis.pdf

[23] A. van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A remote sensing dataset and challenge series," 2018, *arXiv:1807.01232*.

[24] X. Jiang et al., "RoadFormer: Pyramidal deformable vision transformers for road network extraction with remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 102987.

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[25] X. Lu, Y. Zhong, Z. Zheng, and L. Zhang, "GAMSNet: Globally aware road detection network with multi-scale residual learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 340–352, May 2021.

[26] R. Lian, W. Wang, N. Mustafa, and L. Huang, "Road extraction methods in high-resolution remote sensing images: A comprehensive review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5489–5507, 2020.

[27] T. Sun, Z. Di, P. Che, C. Liu, and Y. Wang, "Leveraging crowdsourced GPS data for road extraction from aerial imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7501–7510.

[28] A. Abdollahi, B. Pradhan, N. Shukla, S. Chakraborty, and A. Alamri, "Deep learning approaches applied to remote sensing datasets for road extraction: A state-of-the-art review," *Remote Sens.*, vol. 12, no. 9, p. 1444, May 2020.

[29] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," 2016, *arXiv:1606.00915*.

[30] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[31] D. Chen, Y. Zhong, Z. Zheng, A. Ma, and X. Lu, "Urban road mapping based on an end-to-end road vectorization mapping network framework," *ISPRS J. Photogramm. Remote Sens.*, vol. 178, pp. 345–365, Aug. 2021.

[32] G. Máttyus, W. Luo, and R. Urtasun, "DeepRoadMapper: Extracting road topology from aerial images," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 3458–3466.

[33] Z. Xu, Y. Liu, Y. Sun, M. Liu, and L. Wang, "RNGDet++: Road network graph detection by transformer with instance segmentation and multi-scale features enhancement," 2022, *arXiv:2209.10150*.

[34] F. Bastani et al., "RoadTracer: Automatic extraction of road networks from aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4720–4728.

[35] Z. Li, J. D. Wegner, and A. Lucchi, "Topological map extraction from overhead images," 2018, *arXiv:1812.01497*.

[36] L. Castrejón, K. Kundu, R. Urtasun, and S. Fidler, "Annotating object instances with a polygon-RNN," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4485–4493.

[37] Y.-Q. Tan, S.-H. Gao, X.-Y. Li, M.-M. Cheng, and B. Ren, "VecRoad: Point-based iterative graph exploration for road graphs extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8907–8915.

[38] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4.

[39] H. He, D. Yang, S. Wang, S. Wang, and Y. Li, "Road extraction by using atrous spatial pyramid pooling integrated encoder–decoder network and structural similarity loss," *Remote Sens.*, vol. 11, no. 9, p. 1015, Apr. 2019.

[40] Q. Wu, F. Luo, P. Wu, B. Wang, H. Yang, and Y. Wu, "Automatic road extraction from high-resolution remote sensing images using a method based on densely connected spatial feature-enhanced pyramid," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3–17, 2021.

[41] S.-B. Chen, Y.-X. Ji, J. Tang, B. Luo, W.-Q. Wang, and K. Lv, "DBRANet: Road extraction by dual-branch encoder and regional attention decoder," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[42] J. Mei, R.-J. Li, W. Gao, and M.-M. Cheng, "CoANet: Connectivity attention network for road extraction from satellite imagery," *IEEE Trans. Image Process.*, vol. 30, pp. 8540–8552, 2021.

[43] J. Wang et al., "Deep high-resolution representation learning for visual recognition," 2019, *arXiv:1908.07919*.

[44] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," 2018, *arXiv:1807.10221*.

[45] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[46] F. Yu, V. Koltun, and T. Funkhouser, "Dilated residual networks," presented at the *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 636–644.

[47] X. Li, X. Sun, Y. Meng, J. Liang, F. Wu, and J. Li, "Dice loss for data-imbalanced NLP tasks," 2019, *arXiv:1911.02855*.

[48] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8024–8035.

[49] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[50] A. Mosinska, P. Marquez-Neila, M. Kozinski, and P. Fua, "Beyond the pixel-wise loss for topology-aware delineation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3136–3145.

[51] K. Yue, M. Sun, Y. Yuan, F. Zhou, E. Ding, and F. Xu, "Compact generalized non-local network," 2018, *arXiv:1810.13125*.

[52] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980.

[53] S. Woo, J. Park, J.-Y. Lee, and I. So Kweon, "CBAM: Convolutional block attention module," 2018, *arXiv:1807.06521*.

[54] A. Batra, S. Singh, G. Pang, S. Basu, C. V. Jawahar, and M. Paluri, "Improved road connectivity by joint learning of orientation and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10377–10385.

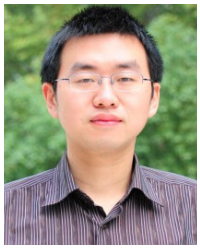
[55] S. He et al., "Sat2Graph: Road graph extraction through graph-tensor encoding," in *Proc. Eur. Conf. Comput. Vis.*, vol. 12369. Cham, Switzerland: Springer, 2020, pp. 51–67.

[56] L. Wang, M. Dai, J. He, and J. Huang, "Regularized primitive graph learning for unified vector mapping," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16771–16780.



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